From Gibbs-Duhem to a local calculation

Guide page 1 | Use with worksheet B/C. Follow the worked C4 example, then attempt C5 independently.

1. Establish the path before using a residual

At fixed T,P, $x\, d\ln\gamma_1+(1-x)d\ln\gamma_2=0$. Differentiate $g=x\ln\gamma_1+(1-x)\ln\gamma_2$: the derivative is L plus that weighted derivative sum, so $dg/dx=L$. In the hand tables g means $g^E/(RT)$, not total mixing Gibbs energy.

On a general path, $\Sigma x_i d\ln\gamma_i=-h^E dT/(RT^2)+v^E dP/(RT)$. Isothermal VLE can have varying pressure; the teaching analysis neglects the excess-volume pressure term. A real isobaric local assessment needs independent h^E. A missing correction is not a failed measurement.

2. The g-first calculation used in C4/C5

Step 1: calculate g at all three compositions, with each row's own x weight. **Step 2:** calculate L at the middle point. **Step 3:** on these equally spaced grids, slope$\approx(g_{right}-g_{left})/(x_{right}-x_{left})$. **Step 4:** subtract center L to obtain r.

C4 value	Clean	Changed upper $\ln\gamma_1=0.36$
g(0.40)	0.48	0.48
g(0.50)	0.50	0.50
g(0.60)	0.48	0.504
L(0.50)	0	0
Slope, then r	0; 0	0.120; +0.120

C4's changed $g(0.60)=0.60\times0.36+0.40\times0.72$. Keep all other values fixed. **Now close the worked table and do C5:** use its new numbers; do not copy C4's answer.

3. What the calculation does not establish

A single finite-difference r is not a dataset-average score. Noise, rounding and spacing affect derivatives. The simple centered formula above assumes uniform spacing; Lab 03 uses a three-point Lagrange derivative with the actual spacing. Three interior points cannot establish a full-range area test.

The clean hand tables come from smooth consistent model functions; the changed entries are deliberately perturbed synthetic values. A nonzero numerical residual indicates a discrepancy under the stated operator and assumptions. It does not uniquely identify an experimental cause.

4. Keep scope labels literal

Not applicable: wrong path for the method. **Not assessed:** missing required information. **Meets/outside this criterion:** a particular eligible numerical test. Avoid collapsing these into a single overall pass/fail label.

For C1/C2, retain signed trapezoids. The Lab's normalized D uses positive and negative area magnitudes and may be undefined near ideality; it is not signed S.

Lab 03: reproduce the demo, then test predictions

Guide page 2 | Worksheet D1-D5, 65-95 min. Watch the real baseline demo, then operate the Lab yourself.

Open Lab 03: Consistency tests

D1. Baseline controls and two views

Set **Data set → Synthetic · isothermal, 350 K**, **Controlled perturbation → Consistent reference**, and **Sampling resolution → 41 points**. The default is 81, so change it explicitly. The synthetic h^E setting has no temperature contribution on this isothermal path.

Under **Inspect**, view **Log-activity ratio & integrated area**, then **Local Gibbs-Duhem residual**. Record D, local score, both statuses and the chart pattern. Expand **Inspect data & numerical results** if more digits are needed. Save the reference; wait until D5 for the fit.

D2-D3. Predict before the perturbation

Write both predictions on worksheet page 3 before selecting either defect. Then change **Controlled perturbation → Canceling errors · zero net area**; set **Amplitude=0.25**. Keep 41 points and the dataset unchanged. Results update automatically. Record D2 and save it.

Change only to **Localized defect near x_1 = 0.35**. Keep amplitude 0.25 and 41 points. Record D3 from both cards and charts. Locate the feature; do not infer the averaged status from the fact that a defect was inserted.

D4. Required sampling comparison

Keep the localized defect and amplitude fixed. Change **Sampling resolution → 161 points**. Record D4 and compare with D3. Note the score changes and peak location/appearance. The setting regenerates sampled synthetic data and changes numerical derivatives; it does not acquire new experimental measurements.

D5. One clean modeling run

Return to **Consistent reference / 41 points**. Click **Run modeling test →** once and wait for **Converged**. Under **Inspect → Modeling residuals**, record Δp (%) and Δy (percentage points). If convergence fails, report no modeling verdict. Save this result before changing inputs.

Use **Your conclusion** and **Export all results**. Rename files clean, canceling, localized-41 and localized-161. You may save a second clean file including the fit. The download is thermodynamic-consistency-study.json. Changing settings clears the previous fit.

Optional reopen check: expand **Study workspace · reopen and compare cases**. Under **Load studies or a comparison JSON**, select your exported file, then choose **Restore** beside its case. Check the restored controls and rerun **Run modeling test →** for a current fit. Imported results are recorded evidence, not a fresh calculation.

Finish E1/E2: an annotated residual plot, a comparison to your predictions, and a claim supported by numbers with a limitation. Take the 10-minute break at study minutes 95-105.

Lab 03: interpret what each method tests

Guide page 3 | Revisit during D5/E. This page explains methods, not the changed-case answers.

Area: a global integral

Let A be area above zero and B the positive magnitude below zero for $L=\ln(\gamma_1/\gamma_2)$. $D=100|A-B|/(A+B)$. Lab 03 integrates line segments and uses D<5% for its isothermal criterion, neglecting the excess-volume pressure term. Opposite local errors can cancel. Missing endpoints mean no full-range verdict unless an explicit extrapolation is assumed; near ideality D may be undefined.

Differential: a local relationship, then an average

$r = d[g^E/(RT)]/dx - \ln(\gamma_1/\gamma_2) + h^E/(RT^2)\ dT/dx$

Lab 03 forms g first, takes three-point finite differences at interior points and reports **100 mean |r|**, with teaching threshold **5**. Isothermal data have no temperature term. Real isobaric data need independent h^E before a corrected verdict. Averages can conceal narrow features; inspect the curve. This is not the fitted Padé/Kojima test implemented in NIST TDE.

Herington: empirical isobaric allowance

$J=150(T_{max}-T_{min})/T_{min}$, using kelvin. The lab uses D-J<10. J describes a boiling-range allowance, not measured h^E. Herington is not applicable to the required isothermal runs. Extrapolated endpoints give conditional results. The full NIST near-ideal decision tree is not implemented.

Van Ness-type: model reproduction

The isothermal fit uses two constant NRTL τ values with α=0.47 fixed. Δp=mean[100|Pcalc-Pobs|/Pobs]; Δy=mean[100|ycalc-yobs|]. Both Δp<1% and Δy<1 percentage point must hold. This teaching parameterization is not NIST TDE's five-parameter model. Small errors show reproduction, not independent predictive validation. Failure can involve model flexibility, property assumptions or data.

Build an inspectable conclusion

Label the chart axes and its case/resolution. Write a specific claim, give two relevant numerical or graphical observations, then state one limitation. Preserve unexpected results and explain a changed prediction. Do not treat resolution as new measurements or a numerical threshold as a universal physical law.

Exports save locally; they do not submit assignments. Use the actual file names/settings so the comparison can be reproduced. The generator is independent of the Lab 08 demonstration data.

Sources: NIST Herington | NIST point test | NIST Van Ness. See the Lab's Sources & scope for implementation limits.

Lab 04: three feeds and an independent energy check

Guide page 4 | F/G, 105-155 min. Predict at slide 34, watch slide 35, then rerun at slide 36.

Open Lab 04: Liquid stability and LLE

Before watching: complete G1 predictions during the 3-minute pause on slide 34. Watch the complete screen demonstration on slide 35. Then use the 25-minute pause on slide 36 to rerun G2-G4 yourself; keep your original predictions and your own calculation record.

1. Fix the model; compare three feeds

Set **Liquid model → Margules · 1 parameter** and **Interaction parameter A → 3**. For **Feed mole fraction z_1**, run 0.02, 0.12 and 0.50. After typing, click **Find equilibrium →**; sliders update immediately. Choose **View → Gibbs energy & common tangent**.

After predicting, the corresponding **Explore a state** presets are a shortcut. Their names disclose the homogeneous-feed labels. Check the A and z values after selection. The hypothetical model has no T/P controls; hold A fixed for this comparison.

2. Record the feed and equilibrium separately

Record **Homogeneous feed**, the final phase label, **Gibbs energy decrease / RT**, and curvature sign. For two liquids record **Coexisting composition x_1^α**, **x_1^β** and **Fraction of β liquid**. β is the component-1-rich liquid, not vapor. For one liquid, report $x_1=z_1$; coexistence fractions do not apply.

Expand **Verify the equilibrium**. Inspect chemical-potential agreement, material balance and full tangent evidence. Positive curvature is local evidence, not proof of global stability. A named unstable or metastable feed can end in a stable two-liquid equilibrium.

3. Do the independent calculations

For $z_1=0.12$ and 0.50, use the actual endpoints in $f\beta=(z_1-x^\alpha)/(x^\beta-x^\alpha)$. Compute fα and reconstruct z_1. Both fractions must lie in [0,1]. Explain what goes wrong if those two endpoints are used for $z_1=0.02$.

For G4 use $m(x)=x \ln x+(1-x)\ln(1-x)+3x(1-x)$. Evaluate m(0.12), then the weighted endpoint energy and their difference. Use natural logarithms. This includes ideal mixing; C used excess Gibbs energy only. Annotate the tangent and compare your result with the Lab energy decrease.

4. Save evidence and close the lesson

Type your reasoning in **Your explanation**, then click **Export study JSON** for each feed. Rename liquid-stability-study.json with z002, z012 or z050. Keep your prediction table, balances, energy calculation and annotated tangent. Small rounding residuals are expected if displayed endpoints are truncated.

Reopen a saved case: expand **Study workspace · reopen and compare cases**, load the study JSON, then choose **Restore** beside its case. Check the restored A and z_1, then click **Find equilibrium →** to calculate again. Imported results remain recorded evidence; restoring inputs does not replace a fresh calculation.

Stop the required investigation at G4. An A sweep is optional within the block, not homework. At 155-170 min attempt H1-H4 before solutions; at 170-180 min complete the existing anonymous Session 3 form, then save all files. No attendance identity is collected.